

## DEM 601 Introduction to Demographic Analysis

Formula sheet

COVERS

lectures 1 to 3

TERM

Fall 2026

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Every equation the course has shown you, in the order it was shown.

### Lecture 1 • Introduction to the Course and Basic Demographic Concepts

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$$P_{n+k} = P_n + \underbrace{{}_k B_n - {}_k D_n}_{\text{natural increase}} + \underbrace{{}_k I_n - {}_k E_n}_{\text{net migration}}$$

population change

What makes a population change

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### Lecture 2 • Demographic Indicators and Person-Years-Lived

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$$\text{indicator} = \frac{\text{numerator}}{\text{denominator}} \times k$$

What an indicator is

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$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

The mean

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$$M = L_0 + \left( \frac{P/2 - \sum f_0}{f_0} \right) i$$

The median of grouped data

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$$\text{ratio} = \frac{a}{b} \times k$$

Ratio

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$$\text{proportion} = \frac{a}{a+b}$$

Proportion

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$$\text{percentage} = \frac{a}{a+b} \times 100$$

Percentage

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$$\text{rate} = \frac{e}{PY} \times k$$

A rate needs time

$$CBR = \frac{\text{births in the year}}{\text{mid-year population}} \times 1000$$

Is it a true rate?

$$IMR = \frac{\text{deaths under age 1}}{\text{live births in the year}} \times 1000$$

Is it a true rate?

$$q = \frac{e}{P_0}$$

A rate is not a probability

$$\bar{P} = \frac{P_0 + P_1}{2}$$

The mid-year population

### Lecture 3 • Population Change

$$\Delta P = P_t - P_0$$

Absolute change

$$\% \Delta P = \frac{P_t - P_0}{P_0} \times 100$$

Percentage change

$$P_n = P_0 + (n \times (r \times P_0))$$

Arithmetic growth

$$r = \frac{P_n - P_0}{n} \div P_0 \times 100$$

Arithmetic growth

$$P_n = P_0 \times (1 + r)^n$$

Geometric growth

$$r = \sqrt[n]{\frac{P_n}{P_0}} - 1$$

Geometric growth

$$P_n = P_0 \times e^{r \times n}$$

Exponential growth

$$r = \frac{\ln(P_n/P_0)}{n}$$

Exponential growth

$$r(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta N(t)}{N(t) \Delta t} = \frac{\frac{dN(t)}{dt}}{N(t)} = \frac{d \ln[N(t)]}{dt}$$

Exponential growth

$$P_n = P_0 \times e^{r \times n}$$

Running the formula forwards

$$2P_0 = P_0 \times e^{r \times n}$$

How long until twice as many?

$$\ln(2) = r \times n$$

How long until twice as many?

$$n = \frac{\ln(2)}{r}$$

How long until twice as many?

**Population Turnover = Losses + Gross Growth**

Standing still is work